

Group Project Guideline

1. Group Formation

- Students are required to form a group of **5–6 members**.
- Group members may be selected at your own discretion.

2. Topic Selection

- **Project Scope & Minimum Requirements:** To ensure a high-quality database design, students should meet these technical benchmarks to complete their projects:
 - **ER Diagram (ERD):** A complete visual representation of the database schema, including entities, attributes, and relationships (One-to-One, One-to-Many, or Many-to-Many).
 - **Normalization:** The database must be normalized to at least **3rd Normal Form (3NF)** to reduce data redundancy.
 - **Tables & Constraints:** Minimum of **6–8 interconnected tables**.
 - Proper use of **Primary Keys** and **Foreign Keys**.
 - Implementation of constraints like NOT NULL, UNIQUE, and CHECK.
 - **CRUD Operations:** Students must demonstrate the ability to **Create, Read, Update, and Delete** data using SQL.
 - **Complex Queries:** At least 5 advanced SQL queries involving JOINS, GROUP BY, HAVING, and subqueries for reporting.
- **Additional Project Idea**

Railway Management System

Description: This system allows users to navigate the complexities of travel by booking tickets, checking seat availability, and managing train schedules across various stations. It focuses on the real-time logistics of transportation and fare calculation.

Focus Tables: Trains, Stations, Routes, Schedules, Tickets, Passengers, Payments, Staff

Library Management System

Description: This project moves away from "messy bookshelves" to a digital catalog where users can search for books, manage due dates, and track fines for late returns. It emphasizes the lifecycle of a physical asset being borrowed and returned.

Focus Tables: Books, Authors, Categories, Members, Loans (Borrow/Return), Reservations, Fines, Staff

Inventory Control System

Description: Designed for businesses to prevent "running out of stock" or "collecting dust". It tracks stock levels, monitors supplier orders, and provides alerts for low inventory to ensure resources aren't wasted.

Focus Tables: Products, Categories, Suppliers, Purchase_Orders, Sales_Orders, Inventory_Logs, Users_Roles

Hospital Management System (HMS)

Description: Beyond just doctors and patients, this system focuses on the organization of healthcare delivery, including appointment scheduling, medical history tracking, and billing.

Focus Tables: Patients, Doctors, Appointments, Departments, Medical_Records, Lab_Tests, Billing/Invoices, Staff

Online Job Portal

Description: A platform that bridges the gap between job seekers and employers. It handles resume uploads, job categorization, and the application process to simplify recruitment.

Focus Tables: Users (Seekers), Companies, Job_Postings, Categories, Applications, Resumes, Employer_Reviews

Classroom & Room Management System

Description: Essential for educational institutions to schedule classes and assign rooms without overlaps or "scheduling headaches".

Focus Tables: Classrooms, Courses, Instructors, Class_Schedules, Enrollments, Resources (Projectors, etc.), Room_Bookings.

Hotel Reservation System

Description: A "digital front desk" that allows guests to browse available rooms, make bookings, and leave reviews for their stay.

Focus Tables: Rooms, Room_Types, Guests, Reservations, Payments, Reviews, Hotel_Staff, Special_Requests.

Pharmacy & Prescription System

Description: A specialized inventory system that links medical prescriptions to patient records. It ensures that pharmacists can track batch numbers and expiration dates for safety and legal compliance.

Focus Tables: Medications, Patients, Doctors, Prescriptions, Inventory_Stock, Suppliers, Sales_Transactions.

Gym & Fitness Membership System

Description: Focuses on subscription management and service scheduling. It tracks different membership tiers (e.g., Monthly vs. Yearly) and coordinates trainer schedules with member attendance.

Focus Tables: Members, Membership_Plans, Trainers, Classes, Class_Schedules, Attendance_Logs, Payments.

Car Rental & Fleet Portal

Description: Manages a fleet of vehicles available for rent. The system must track the condition of the cars, rental durations, customer insurance details, and maintenance history.

Focus Tables: Vehicles, Vehicle_Types, Customers, Rentals, Maintenance_Logs, Payments, Agencies

- Custom Topic & Propose Guidelines

Students may propose their own topic provided they meet the following criteria:

- a. **Complexity Check:** The proposed system must have at least 7 distinct entities.
- b. **Normalization:** They must prove the schema can reach 3NF.
- c. **Use Case:** The student must describe 3 specific "Business Rules" (e.g., "A student cannot borrow more than 5 books at once").

3. Project Requirements

For the selected database system, each group must complete the following tasks:

- a. **Conceptual Study of the Topic**
 - Analyze the general idea of the proposed system.
 - Explain the business logic, objectives, and real-world application.
- b. **System Description**
 - Provide a clear and structured description of the system.
 - Define the main entities, users, and processes involved.
- c. **Entity-Relationship (ER) Diagram**
 - Design and present a complete ER diagram.
 - Clearly identify entities, attributes, primary keys, and relationships.
- d. **Relational Model**
 - Convert the ER diagram into a relational schema.
 - Clearly indicate primary keys and foreign keys.
- e. **Data Definition Language (DDL)**
 - Write SQL statements to create the database structure (tables, constraints, keys, etc.).
- f. **Sample Data Insertion**
 - Insert representative sample data into all tables.
- g. **Data Manipulation Language (DML)**
 - Demonstrate data operations such as INSERT, UPDATE, DELETE, and SELECT queries.
 - Provide meaningful examples that reflect real-world usage of the system.
- h. **Project Report**
 - Compile all components into a structured academic report. Below is the sample structure.
 - Introduction
 - Project description

- Objective
- Database Analysis
 - User/Project Requirements
 - ER Diagram Relational model
- Database Implementation
 - DDL Data Dictionaries (desc table)
 - DML (Insert, Update, Delete)
 - DQL
 - Views
 - Store Procedures and Functions
- Conclusion and Future work
 - Outcome/completed
 - Strong/Weak point
 - Challenges/Difficulty
 - Future work Conclusion

i. **Slide Presentation**

- Prepare a clear and professional presentation summarizing the project.

4. Submission

All project materials must be submitted through the **Learning Management System (Ms Team)**.

5. Project Timeline

- **Week 8** – Submit ER Diagram, Relational Model, and DDL
- **Week 9** – Complete Sample Data and DML Implementation, Data Manipulation Implementation
- **Week 10** – Final Project Presentation and Submission in **Ms Team**

6. Important Note

Students are strongly advised to prepare the **report and presentation slides progressively** throughout the development process, rather than waiting until the final week.

7. Recommended Report Structure

The final report should include the following sections:

- a. **Cover Page**
- b. **Introduction**
- c. **System Description**
- d. **ER Diagram**
- e. **Relational Model**
- f. **Data Definition Language (DDL)**
- g. **Data Manipulation Language (DML)**
- h. **Conclusion**